Tables for technology qualification research

Table I. Technology ordering across qualification constructs

Task and contrast A−B	Construct	A	B	Gap pp	Difference range pp
Python−PyTorch, 105 ads	Mention M	73	83	−9.5	−9.5 to −9.5
	Qualification context H	67	79	−11.4	−11.4 to −11.4
	Unconditional Q	42	22	+19.0	+13.3 to +35.2
	Q plus all recorded examples	43	65	−21.0	−24.8 to −4.8
	Q plus choices/all recorded examples	50	75	−23.8	−27.6 to −7.6
C++−Linux, 87 ads	Mention M	80	52	+32.2	+32.2 to +32.2
	Qualification context H	74	50	+27.6	+27.6 to +27.6
	Unconditional Q	38	47	−10.3	−11.5 to +11.5
Altium−EMC/SI/PI, 56 ads	Mention M	24	36	−21.4	−21.4 to −21.4
	Qualification context H	21	20	+1.8	+1.8 to +1.8
	Unconditional Q	5	15	−17.9	−17.9 to −17.9

Notes: H records any known GENERAL or REQUIRED qualification branch and excludes the separate PREFERRED metric. Q reports confirmed unconditional technology-specific statements, with recorded U states assigned only in the displayed range. “All recorded examples” is shorthand for the example-admissible relation set and includes two unconditional-or-example ambiguities that are positive under either admitted reading. Scope rows are different constructs, not recodings of Q. A separate implementation reproduced the construction and aggregates.

Table II. Construct contrasts where both targets are mentioned

Contrast	Ads	Construct	A	B	Gap pp	Difference range pp
Python−PyTorch	66	H	60	62	−3.0	−3.0 to −3.0
		Q	37	18	+28.8	+21.2 to +51.5
		Q plus all recorded examples	38	51	−19.7	−24.2 to +3.0
		Q plus choices/all recorded examples	45	59	−21.2	−25.8 to +1.5
C++−Linux	48	H	48	46	+4.2	+4.2 to +4.2
		Q	26	44	−37.5	−39.6 to −4.2
		Q plus choices/all recorded examples	32	45	−27.1	−29.2 to +6.3
Altium−EMC/SI/PI	17	H	15	8	+41.2	+41.2 to +41.2
		Q	5	5	0.0	0.0 to 0.0
		Q plus examples	6	8	−11.8	−11.8 to −11.8

Notes: Both targets are mentioned by construction. H can include choice, example, conditional or unresolved branches within GENERAL or REQUIRED. The separate PREFERRED metric is excluded. “All recorded examples” denotes the example-admissible relation set defined in Table I's notes. A separate implementation reproduced the construction and aggregates.

Table III. Narrow-Q firm comparison fragility

Minimum N	Pairs	Supported retention	Supported reversal	Conditional reversal overlap	Conditional retention overlap	Conditional tie overlap	One edit from loss	≤2 edits from loss
10	3	2	0	1	0	0	0	0
5	32	22	5	3	1	1	10 of 27	18 of 27

Notes: N=10 pairs recur at N=5. Distances are sensitivity calculations, not detected errors.